

**STATEMENT OF WORK
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- 1.01. GENERAL: The Architect-Engineer (A-E) shall perform the services as described in the subsequent Sections of this Statement of Work (SOW) and as described below. If there are conflicts between other sections of the SOW and this Section, the provisions of this Section shall have precedence.
- 1.02. BACKGROUND: The existing upper digester compressors 1&2 within Eielson's Wastewater Treatment Plant (WWTP) are currently inoperable. The compressors provide air into the digester, causing the sludge to circulate and assisting in the breakdown process. This helps to promote the breakdown of waste and increase gas production and retention time. Without the compressors operating, sludge is not being properly digested, methane is not being off gassed as designed, and the methane stack tends to freeze up in the winter due to lack of use.

The WWTP staff have reached out to vendors who indicate the old compressors are not manufactured anymore and may require a replacement with blowers. Wastewater permitting may be required depending on the replacement needed. The permitting process will not be the responsibility of the A-E other than to incorporate any changes required by ADEC.

A HAZMAT survey was recently conducted in 2024 for the buildings within the WWTP footprint. The report has been provided as attachment 1 to this SOW. It is important to note that the report inaccurately labels some samples from B2317 as B2316, but the location map provided in Appendix D of the report is accurate. Additionally, this was conducted in relation to another project and included limited testing within this project footprint.

1.03. SCOPE OF DESIGN PROJECT:

- A. General: The work included in the scope of A-E Title I Services (Design Phase Services) for this design project encompasses design of the replacement of the upper digester compressors 1 & 2; assessment of the existing compressors, piping, and all related appurtenances and devices; collecting/reviewing project data and records; evaluation of alternatives; developing design concepts and cost estimates; attendance and participation in review meetings; and making revisions requested by the Government. The contractor is expected to submit intermediate design documents and ultimately produce a complete design based on specified requirements and comments received during the 65%, 95%, and 100% design review meetings.

This information will be utilized to prepare comprehensive drawings, schedules, specifications, and cost estimates for all facets of work and disciplines/trades involved in the construction project. The primary objective is to develop a biddable set of construction documents tailored to address the unique challenges presented by the extreme Arctic conditions at Eielson AFB, AK. The comprehensive documentation will guide the successful execution of the following key design tasks:

B. Specific Project Goals:

1. Design of Compressor Replacement: Replace the existing motor and compressor assemblies. Based on conversations with the manufacturers, it is likely that the compressors will need to be replaced with new blower assemblies.
2. Design shall include:
 - Upgrading motor conductors to larger/proper gauge as required – if design dictates a higher amperage needed to meet the requirement.
 - Re-routing of 3" stainless steel piping as required to accommodate the

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- new blower assembly configuration.
 - Confirming whether the existing explosion proof compressor control panel and switches can be reused. Include reconnection of existing components or replacement with new components if reuse is not feasible.
 - Incorporating the new compressors/blowers into existing SCADA system. Evaluate whether existing PLCs, input sensors, and output devices can be re-used; provide replacement with new components if reuse is not possible.
- C. The work required under this contract includes analysis and design services necessary to design a construction project to accomplish the goals described above.
- D. The final design product shall include the following items: Complete specifications and drawings necessary for the Government to advertise and award a construction contract for the work. The A-E's design shall include provisions for the abatement and disposal of any hazardous materials that would be disturbed as a consequence of performing the work on this project.
- E. The A-E is expected to include drawings and photographs with the body of each submittal as necessary to adequately describe the work involved. Drawings associated with each submittal shall be of sufficient size so that details and notes are easily readable. Drawings 11"x17" or smaller may be bound with the report.

1.04. HAZARDOUS MATERIALS: The following guidance shall be used for dealing with hazardous materials:

- A. Government-Furnished Information: Attachment 1 of this SOW is the "WWTP Hazardous Material Assessment, Nov 24". This report shall be used as the basis for design concerning asbestos and lead-based paint (LBP).
- B. A-E Responsibility and Assumptions:
 - 1. PCB Testing: The A-E shall be responsible for performing testing for Polychlorinated Biphenyls (PCBs) on painted surfaces and concrete that will be disturbed by the project's scope of work. Testing shall, at a minimum, meet the guidelines provided in the EPA Standard Operating Procedure for Sampling Porous Surfaces for Polychlorinated Biphenyls (PCBs) which has been included as an attachment.
 - 2. Asbestos Assumption: For other materials, no additional testing is required. Where materials are suspected of containing asbestos, (e.g., gaskets and valve packing on equipment to be removed) as noted in the Government-Furnished report or discovered during site investigation, but were not tested, the A-E shall assume them to be asbestos-containing material (ACM) for the purposes of the design.
 - 3. Lead: The A-E shall rely on the Government-Furnished report for lead-based paint, which found no lead-based paint. No additional LBP testing is required. The A-E shall direct the contractor to recycle all metal (otherwise not assumed to be ACM) and inform the recycler of the potential presence of lead.
- C. Design and Abatement: The basis for the abatement design shall be the results of the A-E's PCB testing and the assumptions made regarding ACM. The A-E shall prepare all project documents for the remediation of these materials IAW OSHA, EPA, and ADEC regulations.

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- D. Special Demolition Procedures: When construction requires special demolition procedures to protect personnel from exposure to excessive levels of hazardous materials, these requirements shall be specified and coordinated with the Eielson AFB project engineer and the Eielson AFB Environmental Element of the Installation Management Flight. These procedures shall not be less stringent than Federal OSHA and EPA requirements.
- E. Project Document Requirements: The following items are a minimum of what shall be included in the project documents:
4. Project drawings shall denote the precise location, type, and quantity of identified PCBs and all assumed asbestos-containing materials.
 5. A copy of the A-E's final PCB testing report shall be included in the specifications.
 6. Drawings shall note the location of approved disposal areas and specify haul routes for the hazardous materials to be remediated.
 7. All construction contractor notification and permitting requirement shall be specified.

- 1.05 **TASKS:** The design shall be divided into three (3) tasks. The Government shall neither approve payment for a task nor issue Notice to Proceed for the subsequent task until all work under the task is complete and the A-E has provided all the listed Contract Deliverable Items applicable to the task.

Review conferences shall be scheduled by the Contracting Officer (CO) and be held at Eielson AFB. The A-E firm shall attend with all necessary engineering disciplines. Reviewers will submit written comments on the investigation or design. Comments will be provided to the A-E for response prior to the meeting. The A-E may make a presentation at each meeting if necessary for the design or review process. The A-E shall record the proceedings of each meeting and submit a written summary report (minutes) of those proceedings to the CO for written approval. Draft minutes for each meeting, including the responses to each reviewer's comments, shall be submitted to the CO within three (3) working days following each meeting.

All contract deliverable items required for each task shall be submitted to the CO at the same time, except for the draft review conference minutes. Approval from the CO shall be obtained prior to the submission of contract deliverable items separately.

TASK 1: Produce the required intermediate 65% design documents; submitting the required documents, discussing these documents at the 65% Review Conference, and resubmitting a revised 65% design submittal based on the original 65% design submittal and discussions during the 65% Review Conference.

- A. **65% Review Conference:** The conference for discussing the 65% design shall be held following receipt of the 65% design submittal.
- B. **Contract Deliverable Items:** The following items are deliverable as part of Task 1:
8. An electronic copy via DoD Safe of the required 65% design documents to include drawings, draft specifications, cost estimates, design analysis, draft DD1354, *Transfer and Acceptance of Real Property*, and approved pre-design meeting minutes, including all comments and responses. Drawings shall be 22" x 34", and all details shall be legible.
 9. An electronic copy via DoD Safe of the draft Basis of Estimate (BOE) explaining methodologies and assumptions used in developing the cost estimate. The BOE shall include, at minimum, assumed unit costs and material pricing sources; labor rates and

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productivity assumptions; quantity take-off methodologies; escalation assumptions; contingency percentages (if applicable); sources of historical cost data (e.g., RSMeans, proprietary databases); clarification of any exclusions or scope assumptions. A template is provided as attachment 3 of this SOW.

10. The government review period will not begin until ALL deliverable items have been received.

TASK 2: Produce the required 95% design documents; incorporating changes requested in approved 65% review comments; submitting the required documents; and presenting these documents at the 95% Review Conference.

- A. **Draft Final (95%) Cost Estimate:** The 95% design cost estimates provided by the A-E shall have zero (0) contingency.
- B. **Final Review Conference:** The final review conference shall be held following receipt of the 95% design submittal.
- C. **Contract Deliverable Items:** The following items are deliverable as part of Task 2:
 1. An electronic copy of the draft 65% design review conference minutes.
 2. An electronic copy via DoD Safe of the required 95% design documents to include drawings, draft specifications, cost estimates, draft DD1354, *Transfer and Acceptance of Real Property*, and design analysis, including all comments and responses. Drawings shall be 22" x 34", and all details shall be legible.
 3. The Government review period will not begin until ALL deliverable items have been received.

TASK 3: Produce final (100%) design documents as defined in Section 3 of this SOW.

- A. **Final (100%) Documents:** The A-E shall incorporate into the 100% project documents the revisions requested at the 95% review meeting. The final documents shall be delivered to the CO within fourteen (14) calendar days after the final review conference. Each plan sheet shall bear the seal of a Registered Professional Engineer or Architect as applicable. A-E shall provide a completed Submittal Register.
- B. **Contract Deliverable Items:** The following items are deliverable as part of Task 3:
 1. An electronic copy of the draft 95% design review conference minutes.
 2. Two (2) paper copies of the 100% drawings (one (1) 22" x 34" and one (1) 11"x17") and specifications, and an electronic copy via DoD Safe of the following: the 100% drawings and specifications, the design analysis, the construction cost estimate, and the Submittal Register.
 3. An electronic copy of the draft DD Form 1354, *Transfer and Acceptance of Real Property* (DD1354). Note "draft" refers to the stage of the DD1354 as dictated by Real Property (i.e., Draft, Interim, and Final). The draft DD1354 is provided to the construction contractor as the framework for the interim and final DD1354.

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4. An electronic copy via DoD Safe of the complete project specifications, final Submittal Register, project drawings, final design analysis, and the final construction cost estimate. The project specifications shall be delivered in both SpecsIntact and PDF formats. The Submittal Register shall be delivered in Excel and PDF formats. The project drawings shall be delivered as AutoCAD files and PDF format. The cost estimate shall be submitted in the following formats: A PDF of the cost estimate report, a native file in either PACES or MII, and an Excel file. The design analysis shall be in PDF format.
5. An electronic copy via DoD Safe of the complete BOE explaining methodologies and assumptions used in developing the cost estimate.
6. The submission will not be considered complete until ALL required electronic and paper copies have been delivered.

NOTE: Specifications shall be bound and shall be printed on double-sided 8-1/2 x 11 bond paper. Design analysis and cost estimate shall be printed on 8-1/2 x 11 bond paper and shall be bound. All pages shall be numbered consecutively. Provide all plot style files, font files and cross references utilized with the drawing files. All cross references shall be bound to the appropriate drawing files.

- 1.06 REAL PROPERTY DOCUMENTATION (DD FORM 1354): The A-E shall prepare and submit a draft DD1354, *Transfer and Acceptance of DoD Real Property*, as a design support document to facilitate real property accountability, planning, and project close out.

The draft DD1354 shall be developed based on the approved project scope, design documents, and construction cost estimates and shall be coordinated with the Government during the design phase.

The draft DD1354 shall be submitted at the following design milestones:

- 65% design submittal
- 100% final design submittal

At a minimum, the draft DD1354 shall include:

- Project identification data consistent with the approved DD form 1391
- Facility category codes (CATCODEs) and asset descriptions
- Identification of new construction, repair, alteration, and demolition scope
- Estimated construction costs and total project costs
- Estimated capitalization values by facility and/or asset
- Supporting assumptions and calculations used to derive asset values

The A-E shall update the draft DD1354 as necessary to reflect design changes in Government review comments through final design.

- 1.07 SCHEDULE: The A-E shall provide both pdf and native Microsoft Project files (.mpp) after project award for the design schedule with key milestones shown. An updated schedule shall be provided after each Notice to Proceed or at any point a deliverable date has shifted.

Government reviews are fixed as identified in this section. The A-E shall incorporate Government review comments and resubmit revised documents within timeframes directed by the Contracting Officer. Schedule impacts resulting from Government-directed scope changes or additional review cycles shall be addressed

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by contract modification, if applicable. The design schedule shall commence upon receipt of the initial Notice to Proceed (NTP).

1. 65% Design Due: 45 calendar days after receipt of Notice to Proceed to 65%.
2. 65% Gov't Review Period: 21 calendar days after submittal of the 65% Design documents.
3. 95% Design Due: 24 calendar days after receipt of Notice to Proceed to 95%.
4. 95% Gov't Review Period: 21 calendar days after submittal of the 95% Design documents.
5. 100% Document Preparation: 14 calendar days after receipt of Notice to Proceed to 100%.

END OF SECTION

SECTION 2
GENERAL A-E DESIGN SERVICE REQUIREMENTS FOR
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- 2.01 GENERAL SCOPE: The A-E shall perform professional architectural and engineering services such as planning and design, provide labor and materials to accomplish the work specified in this contract. The A-E shall be responsible for all travel and lodging arrangements and expenses involved in the performance of work under this contract. The A-E shall furnish the Air Force with complete and usable products to include design analysis, project drawings, specifications, equipment layouts, cost estimates, submittal registers, pricing schedule amendments (resulting from design omissions), construction contract modifications (resulting from design omissions), and other documents required under this contract to provide complete engineering services for Eielson AFB, Alaska.

A. Refer to Section 1 of this SOW for a detailed description of this specific project.

- 2.02 TYPE OF WORK: Work on the project may involve any or all disciplines including electrical engineering; drafting; cost estimating; and word processing.
- 2.03 ADMINISTRATION: The Eielson AFB CO is responsible for the administration of the contract. No change shall be made in the provisions of this contract without written authorization from the CO. The Base Civil Engineer (BCE) monitors the work and services furnished and evaluates the performance of the A-E as the CO's Technical Representative (COTR). The COTR is not a CO and has no authority to Contractually bind the Government or modify the contract.
- 2.04 GENERAL DESIGN CRITERIA AND REFERENCES: The project entails designing functional concepts suitable for the sub-arctic environment in interior Alaska. The design must adhere to various industry standards, including, but not limited to those of the National Fire Protection Agency (NFPA), United Facilities Criteria (UFC), Air Force Manuals (AFMAN), Air Force Instruction (AFI), and Unified Facilities Guide Specifications (UFGS), and many others. Additionally, it must comply with Air Force-specific criteria, such as AFM 88-3 C2 and EM 385-1-1, as well as Occupational Safety and Health Administration (OSHA) and Americans with Disabilities Act (ADA) standards. Various Air Force Engineering Technical Letters (AFETLs) and Installation Facility Standards (IFS) are also relevant to the project, in addition to other pertinent industry standards, State Department of Environmental Conservation, EPA requirement, and federal, DoD, and AF regulations and policies.

All Air Force Manuals, Publications, and Guide Specifications will not necessarily be furnished. The A-E may acquire them from the Defense Printing Service Detachment Office, Standardization Document Order Desk, 700 Robbins Avenue, Philadelphia, PA 19111-5094, or other Federal agencies. The Special Assistance Desk phone number at Defense Printing Service is (215) 697-2664. Many references and standards are available via the following websites: **www.wbdg.org** and **www.afcec.af.mil/library/index.asp**. The A-E may review any Air Force Manual or publication available at the Base Civil Engineering offices.

- 2.05 GOVERNMENT DOCUMENTATION: The Government will provide digital copies of all drawings that might be relevant from Eielson AFB digital drawing vault. The A-E firm may request additional information via the CO for access to the drawing vault. The A-E should not expect Government personnel to perform a drawing search for him. "As-builts", shop drawings, and operation and maintenance manuals that the Government may have are not guaranteed to be either accurate or complete. It shall be the A-E's responsibility to verify any information or data that is given to them by the Government.

The A-E shall perform reasonable field verification of visible and accessible existing conditions necessary to support design development. The Government does not guarantee the accuracy of completeness of record drawings.

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- 2.06 WARRANTY: The A-E shall bring any questions concerning the design to the attention of the Government during the design phase. The A-E shall verify that the final documents have been field checked to eliminate conflicts and interferences. The A-E shall provide interpretation of construction documents to clarify any ambiguities and shall correct any errors or omissions in his design package. The A-E shall not be responsible for concealed, inaccessible, or undocumented conditions unless such conditions are identified during authorized site visits or investigations performed under this contract.
- A. DISCREPANCIES: The A-E shall be responsible for the professional quality, technical accuracy, and the coordination of all design, drawing, specification, and other services furnished by the A-E under the resulting contract. The A-E shall, without additional compensation, correct or revise any errors or deficiencies in design, drawings, specifications, and other services in an agreeable timeframe with the CO and Contractor to avoid construction delays.
 - B. INTERPRETATION: During the construction solicitation and the construction periods of the project, the A-E shall be available, at no additional cost to the Government, to interpret or clarify plans and specifications, should questions arise.
- 2.07 CONCEPT LIMITATIONS: General information and criteria provided to the A-E shall not prevent the A-E from suggesting or using improved methods and designs with functional or cost-saving advantages.
- 2.08 GOVERNMENT FURNISHED MATERIALS: The Government shall furnish the A-E with the following:
- A. A description of the specific requirements for the design of this project. See Section 1 of this SOW.
 - B. An estimated construction cost range.
 - C. Access to existing drawings of the facility as needed to perform design.
 - D. Eielson AFB specific Division I specifications (to be tailored by the A-E for specific project requirements).
 - E. WWTP Hazardous Material Assessment, Nov 2024

END OF SECTION

SECTION 3
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- 3.01 SCOPE OF WORK: The A-E shall perform all services required to prepare and furnish complete construction documents, with addenda as required, for the accomplishment of this contract. Design services provided under the contract shall not include construction means, methods, techniques, sequencing, or safety procedures, which are the sole responsibility of the construction contractor.
- 3.02 DESIGN SERVICES: Design services shall include all aspects of design required to provide a complete and accurate set of project documents to be used for obtaining contract construction services. This includes the following:
- A. Familiarization with the scope of the project, general conditions governing the performance of the work, and the conditions under which the project will be constructed.
 - B. Coordination with Base Civil Engineering and the using agencies to develop functional relationships and detailed requirements.
 - C. Production of the required design documents.
 - D. Preparation of the draft DD1354 to support real property documentation and planning, coordinated with the Government and consistent with the approved design scope and cost estimates.
- 3.03 DESIGN DOCUMENTS: The contract drawings to be furnished shall be well prepared, complete, and prepared IAW the best professional practice to show clearly and concisely the type and extent of work to be performed. The drawings shall clearly distinguish new and existing work. The drawings shall be drawn to appropriate scales and dimensioned completely and accurately. Each drawing shall have a graphic scale. The A-E shall use a standard sheet format and fonts conforming with Eielson AFB Drafting Standards. Lettering shall be large enough to ensure legible half-size reduction. Each discipline's group of drawings shall contain a detailed legend defining all applicable symbols appearing on that discipline's sheets. The legend shall appear on the first sheet of each group. Standard building material indicators, and symbols for architectural items and for mechanical and electrical equipment shall be used to the greatest extent possible.
- A. DESIGN ANALYSIS: The A-E shall prepare a complete and accurate design analysis to support the project design. Each sheet of the design analysis shall be sequentially numbered and shall carry the title of the project, date of analysis, and the designer's name. The analysis shall be bound and prepared (typed or legibly printed) on 8-1/2"x11" bond paper or standard company form, with 1" left and right margins. The design analysis shall cite specific code references, assumptions, calculation sources, data sources, etc. Calculations and data shall be presented in a neat and logical progression. The analysis shall, at a minimum, include: a brief description of the desired project outcome, anticipated construction schedule, description of existing conditions, site descriptions, existing facility features such as age and type of construction, utility location and access, photos as necessary or required, electrical-load calculations, computer modeling, etc. A constructability report shall be incorporated into the 95% design analysis (see para. 3.07). An updated design analysis shall be submitted at each review stage to describe and justify each decision made at that stage. The final design analysis shall present a clear and logical picture of the final design and shall be edited to exclude reference to changes made during the design process.
 - B. PRICE SCHEDULE: The A-E shall provide a detailed proposed pricing schedule in the format

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specified by the Government.

- C. **DRAWINGS:** The A-E shall provide a detailed schedule of drawings in the format specified by the Government. The A-E shall prepare and furnish drawings to show existing site conditions, general layout of facilities, and the extent, type, and detailed requirements of the work to be performed. Full-scale drawing size for Eielson AFB is ANSI D (22" X 34"). Separate drawings shall be prepared to cover all needed phases of architectural, hazmat, civil, structural, mechanical, fire suppression, electrical, etc. work to be accomplished. Detail drawings shall refer to the sheets and drawings to which they apply. Similarly, where details are called out, the callout shall identify the detail tag and the sheet where the detail is located. In addition to hard copy drawings, the A-E shall provide electronic copies of the drawings via DoD Safe in the Government's most recent version of AutoCAD (version 2020 or later) and in read-only Adobe Acrobat Reader (.pdf file) format. Actual number of required copies of drawings to be submitted is described in Section 1 of this SOW.
- D. **SPECIFICATION TECHNICAL PROVISIONS:** The A-E shall prepare detailed specifications for each category of work for each building trade to meet Government standards for a competitive solicitation. Included shall be a list of all Government-Furnished Equipment (GFE). Specifications shall be written utilizing industry and Government standards. Specifications shall conform to the Unified Facilities Guide Specifications (UFGS) system with UFGS-format section number and page numbers centered at each page bottom. The header of each page shall include the project title and project number on the left-hand side. The specification Table of Contents shall indicate which UFGS sections are included. Brand name references in specifications are prohibited unless approved by the CO or mandated by the Government. However, basis of design information shall be incorporated in the drawings. Specifications shall be tailored to the project and be clear; concise; free of typographical, grammatical, and spelling errors; etc. Specifications shall be typed on 8-1/2" x 11" bond paper with 1" left and right margins. In addition, the complete specifications shall be submitted via DoD Safe, in SpecsIntact format. Actual number of required copies is described in Section 1 of this SOW.
- E. **COST ESTIMATE:** The A-E shall prepare detailed construction cost estimates at the 65%, 95%, and 100% design stages. Cost Estimates shall be based on current, verifiable cost data reflective of construction conditions at Eielson AFB, Alaska. Estimates shall be broken down into line items by major construction divisions and include labor, material, equipment, and contractor markups.

Each cost estimate shall be accompanied by a Basis of Estimate (BOE) documenting assumptions, methodologies, quantities, exclusions, escalation factors, and data sources. Cost estimates shall be prepared using PACES or MII and submitted in both native and PDF formats.

Cost estimates and supporting documentation shall be considered CONTROLLED UNCLASSIFIED INFORMATION (CUI) and shall be handled, transmitted, and stored in accordance with DoD and Air Force CUI requirements, including submission via DoD Safe.

The 95% and 100% design cost systems shall include zero (0) contingency unless otherwise directed by the Contracting Officer.

- 3.04 **DESIGN REVIEW:** Design review shall be held at the 65% and 95% design stages. After the review, the A-E shall not proceed to the next design stage until given Notice to Proceed (NTP) in writing to do so by the CO. each review period will adhere to the times outlined in section 1.06 of this SOW. Reviewers will submit written comments on the design that will be provided to the A-E for response. Additional comments

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and questions may be introduced during the review meeting(s).

- A. DESIGN REVIEW MEETING ATTENDANCE: The A-E shall attend design review meetings at Eielson AFB or remotely, take attendance and minutes, and answer review comments. The review meetings shall be scheduled through the CO. The review meetings will be attended by the Government's project engineer, engineering staff, representatives of the reviewing and using agencies, interested parties, and by the A-E's project manager and necessary staff. Reviewers will submit written comments on the design that will be provided to the A-E for response. In addition to the written comments, verbal comments may be provided at the review meeting(s).
- B. 65% REVIEW: The 65% design package shall be submitted to the CO for Government review and approval. Included shall be the 65% design analysis, well-developed drawings, draft specifications, draft DD1354, final calculations, and selection of material types for construction, draft pricing schedule, draft drawing schedule, detailed cost estimate, preliminary Submittal Register, and the approved pre-design review meeting minutes to include the investigation review comments and responses. Drawings should indicate final layouts, details and sections, and initial schedule layouts with some fill-in remaining.
- C. 95% REVIEW: The 95% design package, which should be the A-E's final design package, shall be submitted to the CO for final Government review and acceptance. Included shall be the final design analysis, drawings, drawing schedule, specifications, draft DD1354, calculations, material selection types, pricing schedule, submittal register, cost estimate, construction schedule, and constructability review. The A-E shall have performed and completed internal Quality Assurance reviews of the design and thoroughly proofread the specifications.
- D. 100% PACKAGE: The 100% package shall be the 95% package with those changes necessitated by the accepted comments received at the 95% design review. Included shall be the 100% drawings, drawing schedule, submittal register, synopsis of work, specifications, draft DD1354, design analysis (including calculations, recommended material selections, and constructability report), pricing schedule, cost estimate, and the approved 95% design review meeting minutes to include the 95% comments and responses.

3.05 DOCUMENTS SUBMITTED: The exact number of documents required for each review submittal, and time frames (subject to negotiation) for their development and review are described in Section 1 of this SOW. All the architectural and engineering design, design analysis, specifications, and drawings shall be developed to a uniform level of completion for each review submittal. All review plans and specifications shall be boldly marked on each page "FOR ENGINEERING REVIEW ONLY, % REVIEW, NOT FOR CONSTRUCTION". The cover sheet of each document shall be boldly marked with the date submitted. Document packages shall include:

- A. MINUTES AND REVIEW COMMENTS: The A-E shall submit minutes and the annotated review comments to the CO for written approval, within three (3) working days following each review meeting. Approved minutes and annotated comments shall be submitted as part of the subsequent design submittal.
- B. REVIEW DOCUMENTS: The A-E shall submit the required number of design packages for review to the CO at each review stage. See Section 1, para. 1.05 for specific requirements.
- C. DDAFT DD FORM 1354: The AE shall submit a deep draft DD1354 at the 65% and 100% design submittals as part of the required contract deliverable items. The draft DD1354 shall be submitted electronically via DoD safe in PDF format, and in editable format if requested by the

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Government. Review and acceptance of the draft DD1354 is for completeness and consistency with the design and does not constitute final Real Property acceptance or capitalization approval.

- D. 100% SUBMITTAL: The A-E shall submit the required number of the 100% design documents to the CO as described in Section 1 of this SOW. The submittals must meet the requirements outlined in Section 1, paragraph 1.05, Task 3 of this SOW. Files shall be delivered and labeled clearly with typed project information to include the date and shall be accompanied by a typed index. Each drawing file shall be made to stand alone by binding all X-REFs. Use only standard AutoCAD font files. Change all menus to standard AutoCAD prior to delivery. Include all *.ctb (plot configuration) files. If *.ctb files are not used, include list of pen weights.

Cost Estimate deliverables shall be prepared and submitted IAW Section 3.03 E of this SOW.

- 3.06 CONTRACTOR'S FURNISHED PRODUCT: The A-E shall provide complete project designs to the Government as specified elsewhere in this SOW. The A-E shall provide any product literature and technical data required to support the design. Designs shall be of the highest professional quality and shall be certified by a registered architect, civil engineer, structural engineer, mechanical engineer, fire protection engineer, electrical engineer, or land surveyor, as appropriate. Each final plan sheet shall bear the seal of a Registered Professional Engineer or Architect.
- 3.07 CONSTRUCTABILITY REVIEW: A review of the design for constructability shall be made by the A-E prior to the submittal of the 95% documents. The A-E shall make a site visit during the 95% design phase to determine whether the plans accurately and completely represent existing conditions and to verify that the construction can proceed as designed. The constructability review report shall be incorporated into the 95% design analysis.
- 3.08 DESIGN DISCREPANCIES: The A-E shall be responsible for clarifying any drawings or specifications determined to be unclear to a contractor during the solicitation or construction process. If contract document changes are required due to A-E error or oversight, the A-E shall also be responsible for correcting any discrepancies that become apparent during the solicitation or construction process. The A-E shall promptly, and at no additional cost to the Government, furnish all necessary labor, drawings, specifications, cost estimates, etc., to the CO to enable modification of the construction contract. The A-E shall have the recommended modifications reviewed and certified by a registered architect or engineer in the appropriate discipline.

END OF SECTION

SECTION 4
ATTACHMENTS FOR
FTQW261406; REPLACE UPPER DIGESTER COMPRESSORS 1&2

Attachments:

1. WWTP Hazardous Material Assessment, Nov 24
2. Eielson Division 1 Specifications, various dates
3. Digester As-Builts
4. Basis of Estimate Template
5. EPA Standard Operating Procedure for Sampling Porous Surfaces for Polychlorinated Biphenyls (PCBs)

END OF STATEMENT